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Exercise 4: Launching Web Applications using App Engine Tools

Aim:

To launch and run a web application locally using Google App Engine tools (Flask) and verify it in a browser.

Procedure:

Step 1: Upgrade Flask and Werkzeug

pip install --upgrade flask werkzeug

This upgrades Flask and its dependencies (Werkzeug, blinker, etc.) in the virtual environment.

Step 2: Verify Installed Package Versions

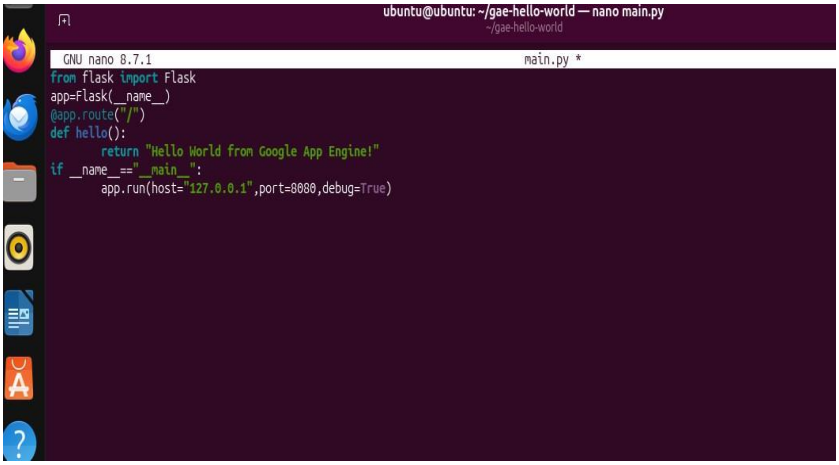
pip show flask
pip show werkzeug

```
(venv) ubuntu@ubuntu:~/gae-hello-world$ pip show flask
Name: Flask
Version: 2.0.1
Summary: A simple framework for building complex web applications.
Home-page: https://palletsprojects.com/p/flask
Author: Armin Ronacher
Author-email: armin.ronacher@active-4.com
License: BSD-3-Clause
Location: /home/ubuntu/gae-hello-world/venv/lib/python3.14/site-packages
Requires: click, itsdangerous, Jinja2, Werkzeug
Required-by:
(venv) ubuntu@ubuntu:~/gae-hello-world$ pip show werkzeug
Name: Werkzeug
Version: 3.1.8
Summary: The comprehensive WSGI web application library.
Home-page:
Author:
Author-email:
License-Expression: BSD-3-Clause
Location: /home/ubuntu/gae-hello-world/venv/lib/python3.14/site-packages
Requires: markupsafe
Required-by: Flask
(venv) ubuntu@ubuntu:~/gae-hello-world$ python main.py
```

Step 3: Review Application Code (main.py)

main.py defines a simple Flask route that returns a Hello World message:

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Step 4: Run the Application

python main.py

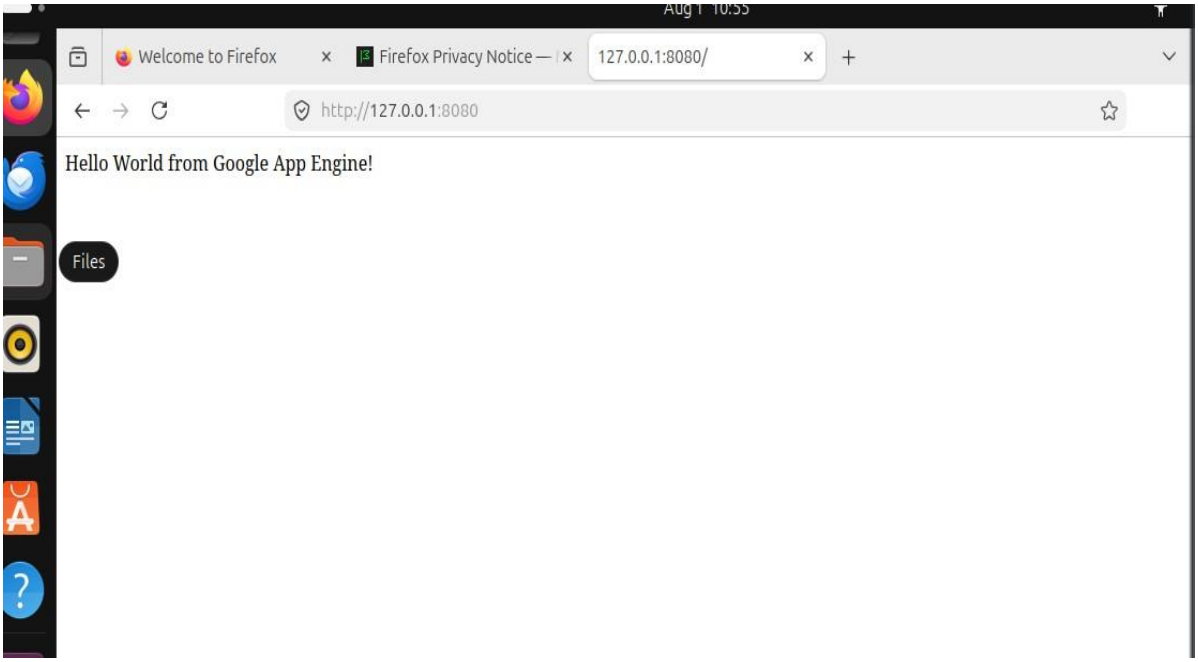
Flask starts the development server on http://127.0.0.1:8080 with debug mode enabled.

```
(venv) ubuntu@ubuntu:~/gae-hello-world$ pip install --upgrade flask werkzeug
Requirement already satisfied: flask in ./venv/lib/python3.14/site-packages (2.0.1)
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: werkzeug in ./venv/lib/python3.14/site-packages (3.1.8)
Collecting blinker>=1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Requirement already satisfied: click>=8.1.3 in ./venv/lib/python3.14/site-packages (from flask) (8.4.2)
Requirement already satisfied: itsdangerous>=2.2.0 in ./venv/lib/python3.14/site-packages (from flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in ./venv/lib/python3.14/site-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in ./venv/lib/python3.14/site-packages (from flask) (3.0.3)
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Installing collected packages: blinker, flask
  Attempting uninstall: flask
    Found existing installation: Flask 2.0.1
    Uninstalling Flask-2.0.1:
      Successfully uninstalled Flask-2.0.1
Successfully installed blinker-1.9.0 flask-3.1.3
(venv) ubuntu@ubuntu:~/gae-hello-world$ python main.py
 * Serving Flask app 'main'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:8080
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 903-333-881
127.0.0.1 - - [01/Aug/2026 10:52:13] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [01/Aug/2026 10:52:13] "GET /favicon.ico HTTP/1.1" 404 -
```

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Output:

The application was accessed from the browser at 127.0.0.1:8080, confirming successful deployment:



Result:

Thus, the Flask-based web application was successfully launched and tested locally using Google App Engine tools.